

THE RELEVANCE OF HARDWARE In Software-Defined World

The current state of the semiconductor industry worldwide is becoming digital, and semiconductors are the underlying technology powering this innovation. The need for exponentially more computing power is being met through innovations, new discoveries, and disruptions. But with increasing use of software-defined equipment, the importance of underlying hardware cannot be ignored

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Nick McKeown (SVP/GM of the Network and Edge Group at Intel)

The traditional fixed-function application-specific integrated circuits (ASICs) hold back network flexibility and prevent the developer from deciding how the network will work. Switch equipment manufacturers have to hard code that functionality into the silicon, which results in generic fixed protocols that aren't suited to every situation. As a result, anytime network administrators need to update their switch protocols, they have to replace the hardware in order to do so.

"The significant jump in performance is due to amazing innovations. We are capable of moving electrons through the channel faster by using increased strain and more low-resistant materials. We are also delivering better energy control through novel high-density patterning

techniques and some streamlined structures, and we're going to be enabling improved power delivery and better routing with higher metal stacks," says Nick McKeown, SVP/GM of the Network and Edge Group at Intel. Hence, it's safe to say that semiconductor innovation and Moore's law are alive.

Now, with technological advancement, we have a programmable switch ASIC and network owners can deploy new protocols as fast as they can develop them through the software layer by just downloading them in the field. As a result, network owners can update their switch functionality to keep pace with new cloud based workloads without having to replace the hardware.

The diagram in Fig. 1 shows the difference between fixed-function ASICs and programmable ASICs. The Tofino pipeline uses the same feed-forward multiple-stage pipeline architecture similar to a fixed-

function ASIC, but with Tofino, all the stages are fully programmable.

The flow of traffic between a fixed and programmable chip may be the same, but when the chip is programmable, the parser deparser and the pipeline stages can be programmed by the user or by the developer. Network architects have total control to decide what features, functions, and protocols are supported within their network, and it all runs at a rate with the same powerful performance and cost.

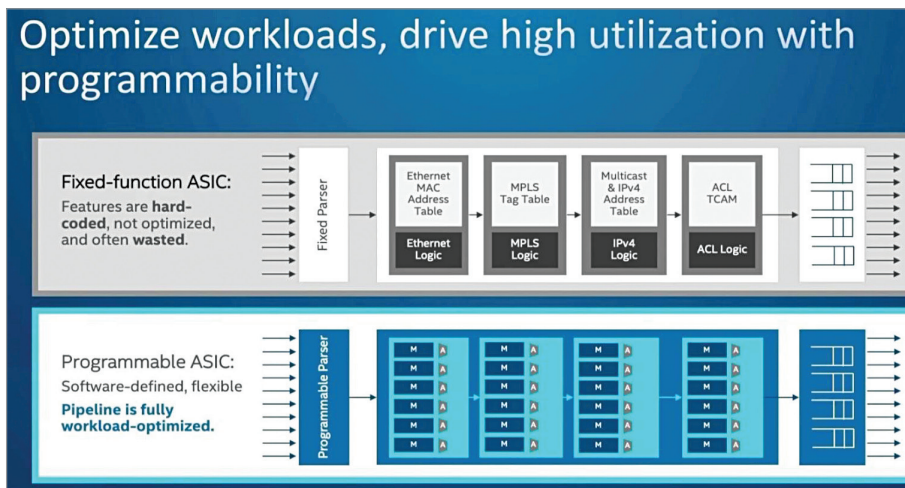


Fig. 1: The difference between fixed-function ASICs and programmable ASICs